

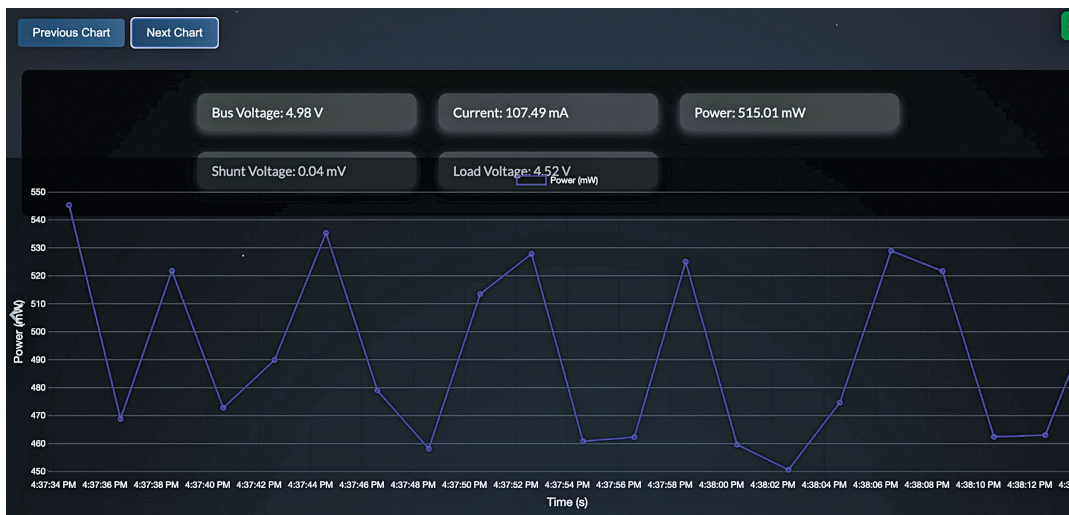


Fig. 6: Satellite module sending data and the webpage hosted by the satellite

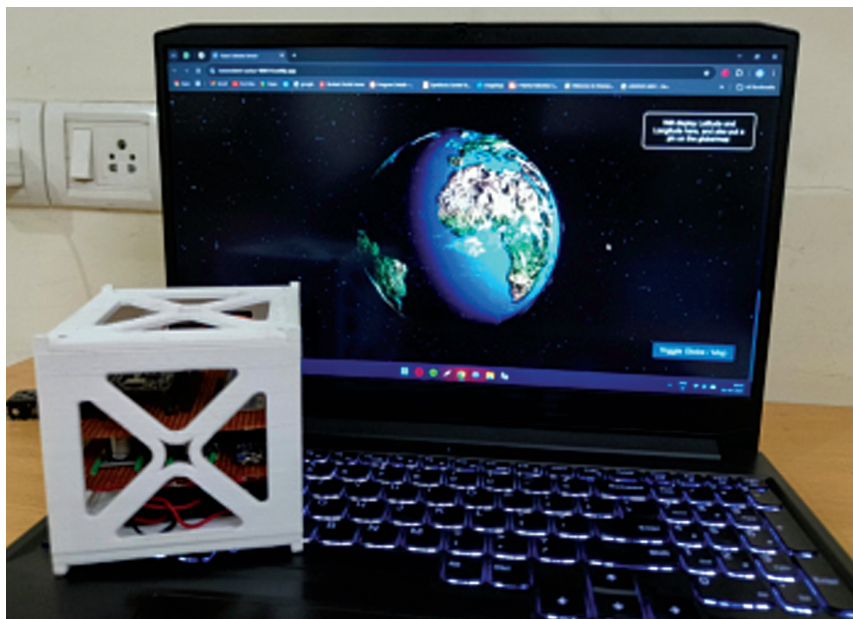


Fig. 7: Final CubeSat appearance with the hosted webpage

To operate the model, the Wi-Fi SSID and password must be configured in the main source code (see Fig. 2). Next, the INA219 sensor library must be installed using the Library Manager. Once the ESP32 board is selected, the code can be uploaded.

Construction and testing

Begin by uploading the main code to the ESP32 board. Once

all components and payloads are connected, assemble and secure them inside the 3D-printed satellite enclosure. Fig. 3 shows the completed circuit inside the 3D-printed structure.

The STL file required for 3D printing the satellite cover can be downloaded from *electronicsforu.com* or accessed by scanning the QR code provided with the article. Fig. 4 illustrates the payload assembly. Fig.

5 shows the fully enclosed CubeSat model.

For testing, the model must first connect to the configured Wi-Fi network. A laptop connected to the same network can then access the hosted webpage to view the real-time data being transmitted. Fig. 6 shows the satellite module sending data to the hosted

webpage. Fig. 7 shows the final CubeSat appearance alongside the webpage interface. **EFY**

TO DOWNLOAD SOURCE CODE AND RESOURCES



Scan this QR code or go to <https://sourcecode.electronicsforu.com> to access and download source code and other resources for this project

Vaibhav Kanade: Mechatronics Engineer and Founder of Pixel Pathways. Serves as Project Lead and Educational Partner, ensuring effective collaboration across the team.

Vrushabh Meghani: Structural Architect of the CubeSat model. Responsible for framework design and integration with the electronic components.

Vedant Ravnang: Handles the digital aspect of the model. Develops the webpage, manages real-time data capture, conducts circuit simulations, and works on advanced PCB design.

Ram Passi: Expert in power systems. Designs robust power modules and circuits, ensuring effective energy management and troubleshooting.

Anjali Aji: Specialist in sensor calibration. Ensures accurate data capture and adaptive responses from the CubeSat model.

Nilanjan Chowdhury: Focuses on system integration and communication between internal modules, enabling smooth data and command exchange across the satellite.